S2 Text. Metric errata, reference corrections and blind verification

HOBench manuscript (AI-agent-run project; see main text). Full records: `results/ERRATA.md`, `families/ERRATA.md`, `families/VERIFICATION.md`, `docs/MUTATION_REPORT.md`.

Metric errata

All five defects were found by trivial baselines or by error analysis (E5 and the aggregation defect by error analysis). Each fix changed the definition for every method in the same way; values before and after are in `results/ERRATA.md`. E1, E2 and E4 were decided before results for panels E, V and P were seen. Entry E3 in `results/ERRATA.md` records reporting changes from design review round 2 (baselines and skill shown with every endpoint; separate control track), not a defect. E5 was triggered by an error analysis of a FastOMA result on panel P.

Errata ID	Defect	Fix
E1	events with an unbounded copy interval and single-gene copies counted as recovered by all-singletons	score only events that both trivial baselines can fail
E2	flat groups were treated as singletons below the root	project each flat group to every level
E4	one big HOG recovered species-level duplications with single-gene copies	headline event recovery over internal branches; leaf events reported separately
E5	membership F1 required families with an unresolved root to form one group	count only separations asserted by the reference
(evaluator)	families sharing a panel-level label were collapsed into one item ($n = 2$ instead of 5 on panel E)	collapse only true collections

Reference corrections after the freeze

- V07 (*X. laevis* homeologs): mapping coverage was stored as a percentage instead of a fraction; two units that shared genes with V03 (Hox) were removed (500 to 498 units). No V results had been seen.
- Y03 (URA1): one quote that did not state a transfer was demoted from T1 to supporting; scored relations did not change.
- Audit notes: NCBI "LOC" strings in V07 unit names were checked and found to be labels, not evidence.

Blind verification

A verifier agent that had not curated the families re-derived min(all, 30) seeded assertions per family from the stored sources, without seeing the claimed values. Two or more discrepancies failed a family; one discrepancy triggered a second sample. Upper bound = two-sided 95% Clopper–Pearson bound on the discrepancy rate.

Family	Items checked	Discrepancies	Upper bound	Verdict
Y01 YGOB pillars (30 units)	49	0	0.073	pass
Y03 URA1	17	0	0.195	pass
K01 ribosomal proteins (30 units)	30	0	0.116	pass
K02 EF-Tu/EF-G	31	0	0.112	pass
K04 analogous enzymes (pooled)	60	0	0.060	pass
E01 kinesins	32	0	0.109	pass
E02 ARF	35	0	0.100	pass
E03 Rab	44	0	0.080	pass
E04 tubulins	31	0	0.112	pass
E05 MCM	32	0	0.109	pass
E06 myosins	31	0	0.112	pass
E07a/b/c adaptins	31 each	0	0.112	pass
V02 opsins	45	1	0.118	pass after resample
V03 Hox	91	0	0.040	pass
V07 *X. laevis* (30 units)	60	0	0.060	pass
P01 plant homeobox	35	0	0.100	pass
P02 bHLH	33	0	0.106	pass
P03 GID1/CXE	32	0	0.109	pass
P04 MIPs	34	0	0.103	pass
V04 Sox	47	0	0.075	pass
V01 globins	37	2	0.182	fail (secondary)

The verifier could not be fully blind in all cases: some sources make the claimed label obvious from the row being checked. The upper bounds show that 30 items bound the error rate only to about 10% per family.